SPOT: Spatio-temporal Pattern Mining and Optimization for Load Consolidation in Freight Transportation Networks

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***Abstract*—Freight consolidation has significant potential to reduce transportation costs and mitigate congestion and pollution. An effective load consolidation plan relies on carefully chosen consolidation points to ensure alignment with existing transportation management processes, such as driver scheduling, personnel planning, and terminal operations. This complexity represents a significant challenge when searching for optimal consolidation strategies. Traditional optimization-based methods provide exact solutions, but their computational complexity makes them impractical for large-scale instances and they fail to leverage historical data. Machine learning-based approaches address these issues but often ignore operational constraints, leading to infeasible consolidation plans.**

This work proposes `SPOT`, an end-to-end approach that integrates the benefits of machine learning (ML) and optimization for load consolidation. The ML component plays a key role in the planning phase by identifying the consolidation points through spatio-temporal clustering and constrained frequent itemset mining, while the optimization selects the most cost-effective feasible consolidation routes for a given operational day. Extensive experiments conducted on industrial load data demonstrate that `SPOT` significantly reduces travel distance and transportation costs (by about 50% on large terminals) compared to the existing industry-standard load planning strategy and a neighborhood-based heuristic. Moreover, the ML component provides valuable tactical-level insights by identifying frequently recurring consolidation opportunities that guide proactive planning. In addition, `SPOT` is computationally efficient and can be easily scaled to accommodate large transportation networks.

***Index Terms*—Logistics, Clustering, Pattern Mining, Optimization, Load Consolidation**

I. Introduction

Freight transportation has grown rapidly over the last decade and has become one of the largest components of the economy, accounting for about 9.0% of the U.S. gross domestic product and 10.3% of the U.S. labor force [1]. Given the vast scale of this industry, efficient planning and operations are highly desirable, not only by logistics companies aiming at increasing profits, but also at the societal level to reduce transportation externalities such as congestion, pollution, noise, and accidents. Academic research [2] and industry practice [3] indicate that a potential strategy to increase profits and mitigate externalities lies in better utilization of container capacity and reduction of partial loads. Partial loads, i.e., shipments that do not fully utilize a container, are often considered unavoidable but remain an important tactic to improve the flexibility of the transportation network and improve customer service satisfaction.

Load consolidation, rooted in the concept of grouping various shipments, parcels, or products into a single batch [4], is considered an effective strategy for reducing the number of partial loads and is widely applied in various logistics contexts, including rail, ground, sea, and air transportation [2], [5]; supply chain network design [6]; and urban congestion challenges [7]. However, transportation networks are highly intricate, requiring extensive coordination of human and material resources, with decisions often decentralized across multiple management units, which makes load consolidation across multiple terminals a complicated task. *Due to the fact that different terminals have different planners who do not have visibility over other terminals in the transportation network, it becomes crucial to define consolidation points well in advance* to ensure alignment between terminals, as well as with other transportation management components, such as driver scheduling, personnel planning, and terminal operations. This complexity makes determining optimal consolidation strategies particularly challenging.

Novel algorithms and techniques designed to address these complexities have been the topic of significant research. Specifically, prior research in [8] examines optimal dispatching rules for *temporal consolidation*, where orders are intentionally held and shipped together, either after a fixed time interval or once a threshold volume is reached. However, these studies typically focus on a single transportation route or shipment path, and do not address the more challenging problem that involves groups of loads or multiple routes. Another research direction [5], [9], [10] formulates load consolidation across multiple origin–destination pairs as a multi-stop pick-and-delivery vehicle routing problem with time windows (m-PD-VRPTW), also known as *vehicle consolidation*. This approach can be viewed as an extension of the vehicle routing problem (VRP), and both mixed-integer programming (MIP) [5], [10] and heuristic methods [5], [9] have been developed based on existing VRP algorithms. Nevertheless, m-PD-VRPTW typically involves a significant computational overhead. The algorithms have been demonstrated primarily on small-scale instances, and are difficult to scale to real-world problems. In addition, [6] consider *terminal consolidation* for middle-mile

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logistics network design, in which loads are routed through predetermined intermediate terminals together for consolidation. These models focus more on the long-term impact of network arrangement and the coordination of other time constraints. The integration of machine learning techniques, i.e., clustering, and association rule mining, is investigated in [2], [11] to evaluate the consolidation performance.

Although considerable progress has been made in load consolidation, existing methods still suffer from at least one of the following challenges:

(C1) - Computational Complexity. The first challenge concerns the high computational complexity inherent to load consolidation problems when multiple origins and destinations are involved. It is well-known that these problems are NP-hard, making it impractical to solve large-scale instances optimally within a reasonable time [5], [6], [9], [10]. Although heuristic algorithms are often employed to mitigate the computational burden of the exact solution methods, their performance can be hampered by the vast search space associated with large-scale instances [5], [6].

(C2) - Restricted Conditions for Operational Consolidation A second challenge lies in the restricted conditions for load consolidation in practice, which are typically addressed in the industry through long-term interactions between the decentralized terminal planning units and other components within transportation management frameworks [12]. A common issue arises when certain terminals cannot accommodate consolidation due to a lack of necessary loading/unloading equipment, space, or personnel. Additionally, certain routes may not support load consolidation because of insufficient transportation frequency or conflicts with current driver schedules. This operational challenge is largely overlooked in the literature. Most proposed algorithms are executed in a greedy and myopic manner [5], [9], [10], assuming that the resulting consolidation routes are operational and ignoring the long-term impact of load consolidation on the broader transportation framework.

(C3): Insufficient Precision in Consolidation Decisions. Although [11] demonstrates that consolidation opportunities exist within load clusters identified based on latitude, longitude, and specific airports or ports, recognizing these opportunities does not guarantee that loads can actually be merged. On any given operational day, factors such as departure times, package sizes, and vehicle capacities play a critical role in determining whether loads can be consolidated. Similarly, [2] highlights the concept of "associated" loads from the pattern mining results, but also ignores the process of constructing feasible consolidation decisions, which involves time and capacity constraints, intermediate terminal selection, and optimal vehicle routing.

(C4) - Limited Use of Historical Data A fourth challenge lies in the limited integration of historical data into current consolidation models. Existing approaches treat each problem instance as completely new, without leveraging past instances and corresponding solutions to enhance efficiency. In reality, logistics systems exhibit spatial and temporal patterns, such as traffic hot spots [13], periodic or predictable demand [14], and recurring trip patterns [15]. The spatial and temporal patterns provide valuable insights into various aspects of logistics, including inventory management [16], monitoring and anomaly detection [17], and delay forecasting and management [15]. However, optimization and heuristic models for load consolidation do not take advantage of these historical data insights.

(C5) - Lack of Comprehensive Testing Datasets The last challenge pertains to the shortage of large-scale, realistic testing datasets in the field. For instance, [9] adapts a vehicle routing problem with time windows dataset to evaluate their proposed algorithm, while [10] relies on a randomly generated network with only five nodes. Although [5] examines heuristic solutions, the largest instance considered involves 58 clients, which does not reflect real-world operational complexity. Without high-quality, large-scale datasets that mirror actual operational environments, it is difficult to validate the robustness and computational efficiency of proposed algorithms.

This paper proposes `SPOT`*, an integrated end-to-end framework for load consolidation that combines Machine Learning (ML) for tactical planning and optimization for real-time operations (see Figure 1) to address these challenges*. During tactical planning, due the decentralized nature of the considered transportation networks, `SPOT` identifies promising *consolidation points*, i.e., locations where load consolidations can take place. `SPOT` uses spatio-temporal clustering [18] on partial loads that share the same destination (left of Figure 1) and then frequent itemset pattern mining [19] to analyze these clusters, identify load groups that frequently appear together in the historical data, and select potential consolidation candidates (middle of Figure 1). During real-time operations, `SPOT` uses these consolidation points inside an optimization model to determine consolidation decisions, using real-time data about scheduled departure times, truck or container utilization, and costs (right of Figure 1).

The contributions of `SPOT` can be summarized as follows:

- `SPOT` is the first integrated framework for real-world load consolidation tasks that integrates machine learning and optimization. `SPOT` spans the entire process from extracting consolidation candidates from historical data for planning purposes to determining cost-saving and operationally feasible consolidation routes for a given operational day.
- The ML module of `SPOT` integrates spatio-temporal (ST) clustering with constrained frequent itemset mining (CFIM) to identify frequent consolidation candidates from historical data. In doing so, requirement ***C4*** is effectively addressed. As highlighted before, isolating consolidation candidates in advance at the tactical level serves as a foundation for coordinating load consolidation between terminals planners and with other components of the transportation management systems, such as driver scheduling, personnel planning, and terminal management.
- For operational consolidation decisions on a specific operational day, `SPOT` uses an optimization model that utilize only the consolidation points identified by the ML module,

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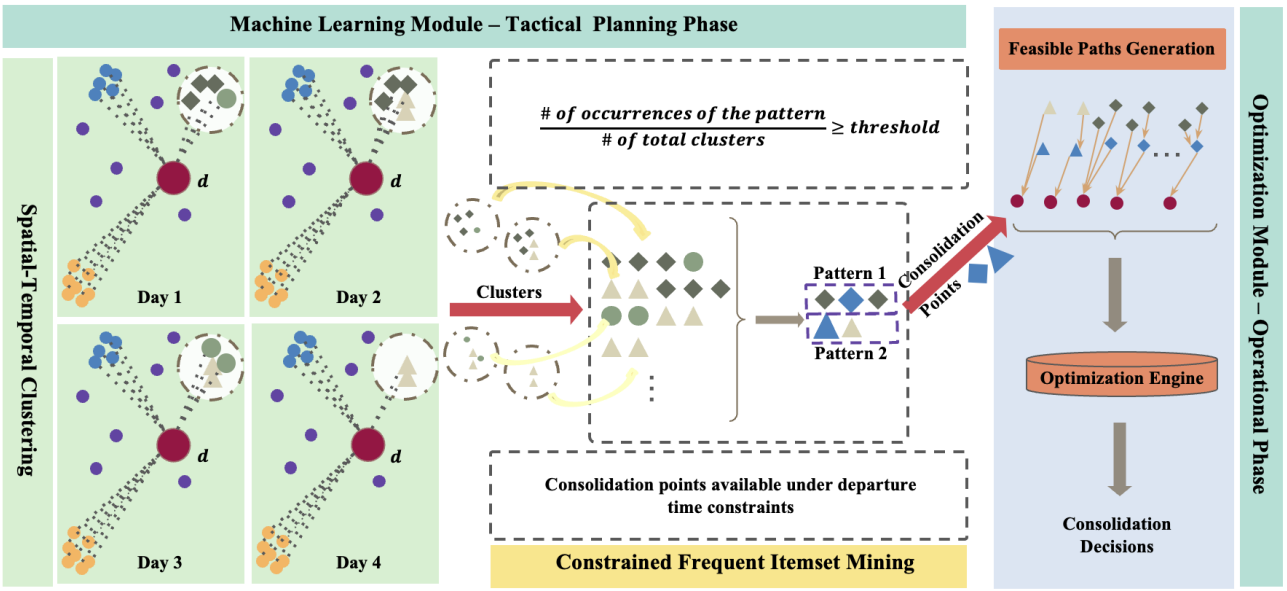


Fig. 1: The Overview of SPOT.

where necessary preparations have been made in advance. The optimization is a mathematical programming model that determines the optimal consolidation route decisions within the context of terminal consolidation. By combining the ML output with the optimization model, the resulting consolidation decisions are both operationally feasible and effective, thereby addressing requirements ***C2*** and ***C3***. In addition, these consolidation decisions are computed independently and in parallel for each destination. This makes it possible to determine consolidation decisions across the entire network efficiently addressing requirement ***C1***.

- SPOT is evaluated on real load data covering the entire U.S. transportation network, addressing requirement ***C5***. The experiments demonstrate the competitive performance of SPOT in terms of travel distance reduction and cost savings, while also offering significant long-term insights.

The rest of this paper is organized as follows. Section II describes the related work. Section III introduces the problem considered in this paper. Section IV describes SPOT, the proposed load consolidation framework. Section V describes the experimental setting and Section VI describes the experimental results. Section VII concludes the paper.

II. Related Work

Load consolidation has been widely studied in both academia and industry. In addition to the literature on load consolidation presented in the introduction, the SPOT approach is also closely related to research involving spatio-temporal (ST) clustering, frequent itemset mining (FIM), and optimization methods in logistics problems, as detailed in the following subsections.

a) Spatio-Temporal (ST) Clustering in Logistics: The large volume of Spatio-Temporal (ST) data generated in recent years, along with developments of geolocation technology (e.g., GPS), has led to a growing interest in ST clustering techniques. These techniques group data points according to latitude, longitude, and an extra time dimension [20]. Although ST clustering is critically important in numerous domains, including image processing and pattern recognition, environmental studies, and etc, there are relatively few studies exploring its potential in decision making for logistics. In contrast, SPOT employs this idea and leverages ST clustering for load consolidation.

b) Frequent Itemset Mining (FIM) in Logistics: Data mining has long been considered a key factor in the success of logistics improvement initiatives, helping to extract valuable insights from various areas, such as supply activity profiles, transportation profiles, and warehouse activity profiles. Specifically, focusing only on frequent itemset mining (FIM), [21] proposes mining patterns of cargo items frequently shipped to military camps, leading to an ontology-like knowledge base for a specialized military transportation network. In addition, [22] extends the concept of an item to include events with temporal validity (e.g., truck travel, loading/unloading, terminal breakdown) to identify frequent co-occurrences in historical data. These event patterns were transformed into predictive rules, providing actionable insights. For example, detecting co-occurring events with adverse outcomes enables managers to take preventive measures. SPOT relies on these insights and treats partial loads in a transportation system as items; this makes FIM a promising approach for pinpointing suitable consolidation candidates from historical data.

c) Optimization in Logisitics: Optimization models have been employed in nearly every aspect of logistics to enhance efficiency, reduce costs, and improve decision-making processes. For instance, [23] formulates an optimization model to determine purchasing periods for oil, aiming at minimiz-

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ing total purchasing and inventory costs. Meanwhile, [24] addresses the storage replenishment routing problem using mixed-integer programming. Recognizing the significant computational overhead of such formulations, [23] and [24] also employ techniques such as variable neighborhood search, MIP-based approximation heuristics and routing-based heuristics to effectively address large-scale instances.

III. Problem Statement

This section specifies the load consolidation problem and the notations used throughout this paper.

a) Freight Transportation Networks: This paper considers a freight transportation network characterized by spatial structure and temporal attributes and represented by a directed graph $G = (V, A)$. The spatial structure comprises approximately 1,000 *terminals* distributed across the United States. The temporal structure is organized around daily sorting periods: each operational day is divided into sorting periods (referred hereafter as *sorts*), typically three to four hours long, during which packages within the loads are processed [25]. Consequently, the inclusion of sorts is crucial for accurately describing the load transportation activities which take place between an origin terminal-sort pair and a destination terminal-sort pair. In this setting, the set of nodes V consists of terminal-sort pairs, and A denotes the set of existing direct routes among these nodes. Formally, each node $v = (v^\sigma, v^\eta)$ is defined by its terminal v^σ and sort v^η. Furthermore, a load arriving at sort v^η must adhere to the latest arrival time to ensure timely processing; its departure time from v^η cannot be earlier that the earliest departure time associated with the sort. These latest arrival time and earliest departure time for sort v are denoted by $arr(v^\eta)$ and $dep(v^\eta)$, respectively.

b) Load Consolidation: A load l in the transportation network is characterized by spatial and temporal attributes as follows:

$$l = (o_l = (o_l^\sigma, o_l^\eta), d_l = (d_l^\sigma, d_l^\eta), t_l, due_l)\,, \tag{1}$$

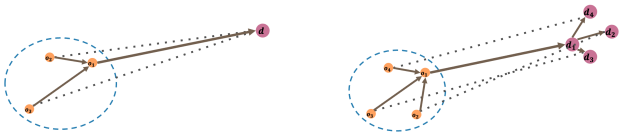
where o_l represents its spatio-temporal origin node, d_l its spatio-temporal destination node, t_l is its scheduled departure time from o_l ($t_l \geq dep(o_l^\eta)$), and due_l is its due date d_l to ensure service. The number of transit days $\omega_l = due_l - day(t_l)$ is the differences (in days) between the due date and scheduled departure time.

A consolidation point is a node (load) where multiple loads can consolidate before traveling together across the network to the common destination. These consolidation points must be identified during tactical planning to synchronize the various terminals in the network that operate largely independently. A **consolidation path** can then be characterized as

$$l = (o_l = (o_l^\sigma, o_l^\eta), h_l = (h_l^\sigma, h_l^\eta), d_l = (d_l^\sigma, d_l^\eta), t_l, t_h, due_l)\,, \tag{2}$$

where $h_l \in V$ is a consolidation point and t_h is the departure time from h_l after its consolidation. This paper makes three assumptions for the load consolidation framework:

(A1) Partial Loads Only: Loads that are already fully utilized are excluded from consolidation, as there is no clear incentive to split or reconfigure a fully utilized load. By contrast, combining multiple partial loads into fewer trailers can substantially reduce the total number of trips and thus lower transportation costs.



(a) Consolidating Loads for a Single Destination.

(b) Consolidating Loads for Multiple Destinations.

Fig. 2: Illustrating Load Consolidations. The dashed lines represent scheduled load transportation, while solid lines indicate modified transportation after consolidation. The dashed circles highlight groups of load origins considered for consolidation.

(A2) Same Due Date, and Destination: Only loads sharing the same destination (d_l) and due date (due_l) can be consolidated (hereafter referred to as "consolidation condition"), as illustrated in the Figure. 2a. The consolidation of loads with multiple destinations that are in close proximity (as shown in Figure. 2b) is a natural extension of SPOT; it is beyond the scope of this work for ease of deployment reasons.

(A3) Consolidation at Existing Origins: The choice of consolidation points is restricted to the existing origins of the considered partial loads. In Equation (2), this means that h_l must already be the origin of another load $\mathfrak{h}$ and that both l and $\mathfrak{h}$ are effectively consolidated together. Again, this assumption is motivated by practical deployment reasons.

These assumptions are primarily motivated by operational feasibility, aiming to ensure promised service levels and minimize disruptions to the scheduled load plan, while still enhancing consolidation wherever possible.

The goal of load consolidation is to find a feasible and cost-effective consolidation plan that

- defines the set of potential consolidation points $H \subset V$ at the technical planning level, using historical records to ensure alignment with driver scheduling, personnel planning, and terminal operations.
- chooses consolidation paths for partial loads using the predefined set of consolidation points to minimize the total transportation cost, subject to constraints on scheduled departure times, planned volume, and trailer capacity.

IV. The SPOT Framework for Load Consolidation

This section describes SPOT, an end-to-end framework that uses machine learning at the tactical planning level and optimization during operations, as shown in Figure 1. This section reviews the details of these components and how they interact. Since SPOT is implemented independently for each destination, the discussion below focuses on a single destination. Note that the decomposition by destination has significant scalability benefits.

A. The Machine Learning Component

The goal of the ML component is to identify potential consolidation points for destination d through spatio-temporal clustering and constrained frequent itemset mining.

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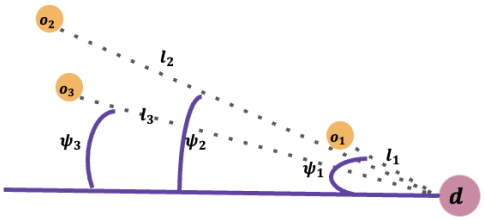


Fig. 3: Consolidation Along the Origin-Destination Path.

1) Spatio-Temporal Clustering (ST Clustering): For a given destination d, the ST clustering receives as input all partial loads in the historical dataset with $d_l = d$. The ST clustering then groups partial loads that are "close" to one another and share the same due date (due_l). For each partial load l, an event data point $\mathbf{x}_l$ is defined to capture the key attributes for clustering:

$$\mathbf{x}_l = (o_l, due_l) \tag{3}$$

where the *spatial component*, $\mathbf{x}_l(s) = o_l^\sigma$, represents the load origin; and the *temporal component*, $\mathbf{x}_l(t) = (o_l^\eta, due_l)$, captures the origin sort and the load due date.

Partial loads are clustered based on their spatial and temporal proximity. While latitude–longitude coordinates (along with Euclidean or Haversine distances) and the absolute difference between timestamps are commonly used for ST clustering [18], it can be misleading in the long-haul load transport context, where consolidation may occur along the route. For example, consider three partial loads l_1 l_2, and l_3 destined for d as shown in Figure 3. A conventional metric based on $Dist(\mathbf{x_1(s)}, \mathbf{x_2(s)})$ and $|(\mathbf{x_1(t)} - \mathbf{x_2(t)})|$ would conclude that the distance between l_1 and l_2 is too large, thus excluding them from the same cluster. However, o_1^σ lies on the route from o_2^σ to d, and thus a modest detour would enable l_2 to consolidate with l_1 if it reaches o_1^σ before t_1.

Polar coordinates capture such potential consolidations. Let $\varphi(\mathbf{x})$ denote the angle between the origin-destination route ($\mathbf{x}(s)$,d) and a reference direction (e.g., West) quantifying the orientation of the load route relative to d. The spatial proximity between two nodes can then be defined as:

$$D_s(\mathbf{x}_1, \mathbf{x}_2) = |\varphi(\mathbf{x}_1) - \varphi(\mathbf{x}_2)|. \tag{4}$$

which measures the difference in route alignment. This formulation ensures that loads with similar directional orientations can be clustered together.

The temporal proximity metric is defined as the difference between the due dates of the partial loads:

$$D_t(\mathbf{x}_1, \mathbf{x}_2) = |due_1 - due_2|. \tag{5}$$

Given the spatio-temporal proximity measures defined above, SPOT uses DBSCAN (Density-Based Spatial Clustering of Applications with Noise) [26] to cluster partial loads. DBSCAN is well-suited for this task because consolidation corresponds to identifying high-density regions where loads exhibit similar spatial and temporal characteristics. Unlike partitioning methods such as k-means or Partitioning Around Medoids (PAM), which require specifying a fixed number of clusters, or hierarchical methods such as BIRCH or Chameleon that focus on nested structures, DBSCAN can detect clusters of arbitrary shape and size based on the defined proximity criteria. Specifically, spatio-temporal nodes are grouped in one cluster if they satisfy the following conditions:

$$D_s(\mathbf{x}_1, \mathbf{x}_2) \leq \epsilon \wedge D_t(\mathbf{x}_1, \mathbf{x}_2) = 0, \tag{6}$$

where ϵ is a pre-defined angle alignment threshold. By enforcing the temporal condition, only loads with the same due date are eligible for consolidation. DBSCAN takes, as input, the set of event data points

$$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, ...\mathbf{x}_{|\mathbf{X}|}\} \tag{7}$$

which represents all partial loads destined to d, and outputs a set of clusters

$$\mathcal{C} = \{C_1, C_2, ..., C_N\} \tag{8}$$

where each C_i is a subset of $\mathbf{X}$.

2) Constrained Frequent Itemset Mining: Once the clusters are identified, SPOT uses Constrained Frequent Itemset Mining (CFIM) to identify loads that are frequently co-occurring for each day of the week (abbreviated as "dow"). This will make it possible to select the consolidation points, which is the ultimate goal of CFIM. As shown in Figure 1, CFIM takes, as input, the set of clusters C for destination d. Each cluster C_i contains a set of partial loads, with each partial load l represented by event data point $\mathbf{x_l}$. CFIM does not use the $\mathbf{x_l}$ data points directly, since they are specific to exact due dates and the goal is to extract repeating patterns. Instead, it uses data points of the form $\tilde{\mathbf{x}}_l$, where the exact due date has been replaced by meaningful features, i.e., its corresponding day-of-week and the number of transit days. The clusters remain the same, but the data has been abstracted to enable the identification of frequent consolidation patterns on a day-of-week basis. The updated representation is defined as

$$\begin{aligned} \tilde{\mathbf{x}}_l &= (o_l, due_l^{dow}, \omega_l), \quad \mathcal{X} = \{\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, ...\tilde{\mathbf{x}}_M\} \\ \tilde{C}_k &= \{\tilde{\mathbf{x}}_l \mid \mathbf{x} \in C_k\}, \quad \tilde{\mathcal{C}} = \{\tilde{C}_1, \ldots, \tilde{C}_N\}. \end{aligned} \tag{9}$$

Intuitively, the goal is to find elements of $\mathcal{X}$ that frequently occur together in the same clusters in $\tilde{\mathcal{C}}$.

Following [27], a *consolidation candidate* (itemset) $S \subset \mathcal{X}$ is deemed *frequent* if its support, i.e., the fraction of clusters containing S, meets or exceeds a pre-defined threshold *min_sup*, where the support of a set S is given by

$$sup(S) = \frac{\sum_{k=1}^{|\tilde{\mathcal{C}}|} \mathbf{1}_{(S \subseteq \tilde{C}_k)}}{|\tilde{\mathcal{C}}|} \tag{10}$$

Any frequent S discovered in this manner indicates that its constituent loads frequently co-occur for d.

The consolidation candidates identified as so are not guaranteed to contain actual consolidation due to temporal constraints. To remedy this limitation, SPOT forces each consolidation candidate to include at least one time-feasible consolidation opportunity. This time-feasibility check filters out infeasible consolidation candidates, thereby retaining only the most useful information for tactical planning and reducing the search space more effectively for the subsequent optimization

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model. In particular, the time-feasibility check, κ, for each candidate S is as follows:

$$\kappa(S) = \exists\ \tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j \in S : \\ dep(o_i^\eta) + \tau(o_i^\sigma, o_j^\sigma) \leq arr(o_j^\eta) + (\omega_i - \omega_j) \vee \\ dep(o_j^\eta) + \tau(o_j^\sigma, o_i^\sigma) \leq arr(o_i^\eta) + (\omega_j - \omega_i). \tag{11}$$

Here o_i^σ and o_j^σ denote the origins of loads i and j, respectively, and $\tau(\cdot,\cdot)$ is the traveling time between two origins; the terms $dep(o^\eta)$ and $arr(o^\eta)$ represent the earliest departure time and latest arrival time of a load with respect to sort o^η. The differences in transit days, i.e., $(\omega_i - \omega_j)$ and $(\omega_j - \omega_i)$, ensure that loads requiring longer transit times can still be consolidated with those having fewer transit days, provided that their routes overlap en route to the common destination. Observe that, whenever an itemset S satisfies κ, so does any superset of S, which implies that κ is a *monotone* constraint. This property is highly desirable in frequent itemset mining.

To efficiently extract these constrained frequent itemsets, SPOT uses the Frequent Pattern Growth (FP-growth) algorithm [28]. Compared to the Apriori algorithm, the FP-growth algorithm is computationally more efficient because it uses a *divide-and-conquer* strategy to mine a compressed FP-tree representation of the dataset. In the constrained version of FP-growth, the feasibility check is incorporated at each resulting pattern of the FP-tree to discard infeasible candidate sets.

Let $\mathfrak{S}$ denote the output of FP-growth, which is the set of consolidation candidates that satisfy the time-feasibility check. For any $S \in \mathfrak{S}$, the corresponding *consolidation points* are defined as

$$H(S) = \{o_l | \tilde{\mathbf{x}}_l \in S,\ \exists\ \tilde{\mathbf{x}}_{l'} \in S : \tilde{\mathbf{x}}_{l'} \neq \tilde{\mathbf{x}}_l \wedge \kappa'(\tilde{\mathbf{x}}_{l'}, \tilde{\mathbf{x}}_l) \text{ is true}\}. \tag{12}$$

where

$$\kappa'(\tilde{\mathbf{x}}_{l'}, \tilde{\mathbf{x}}_l) == (dep(o_{l'}^\eta) + \tau(o_{l'}^\sigma, o_l^\sigma) \leq arr(o_l^\eta) + (\omega_{l'} - \omega_l)). \tag{13}$$

Accordingly, the set of consolidation points for d is then defined as $H = \cup_{S \in \mathfrak{S}} H(S)$. Intuitively, a consolidation point is the origin of a load that can be consolidated with at least one other load. For instance, o_1 in Figure 3 is a consolidation point.

B. The Optimization Component

During real-time operations, there are $\mathfrak{L}$ planned loads destined for d on a specific day, among which L are partial. The set of consolidation candidates $\mathfrak{S}$ and consolidation points H is also available from the tactical planning stage. The optimization module selects the most cost-effective consolidation decisions by leveraging real-time data on scheduled departure times, truck or container utilization, and costs. It starts by identifying feasible consolidation routes for each partial load using $\mathfrak{S}$ and H. SPOT then uses a mathematical model to determine the optimal consolidation routes.

1) Feasible Path Generation: The feasible path generation only considers the subset L_C of loads with a valid consolidation load-pair, i.e.,

$$L_C = \{l \in L \mid \exists\ S \in \mathfrak{S},\ \text{s.t. } \tilde{\mathbf{x}}_l \in S\},$$

The set P_l of all feasible paths for a load $l \in L_C$ contains two types of paths: • *The Direct Route* (o_l, d_l) from origin to destination; • *Consolidation Routes* of the form (o_l, h, d_l), which includes a consolidation point $h \in H$. It is defined as

$$P_l = \{(o_l, d_l)\} \cup \{(o_l, o_\mathfrak{h}, d_l) \mid \\ \mathfrak{h} \in L_C\ \&\ o_\mathfrak{h} \in H\ \&\ d_\mathfrak{h} = d_l \\ \&\ due_l = due_\mathfrak{h}\ \&\ \kappa''(\tilde{\mathbf{x}}_l, \tilde{\mathbf{x}}_\mathfrak{h}) \text{ is true}\} \tag{14}$$

where $\kappa''(\tilde{\mathbf{x}}_l, \tilde{\mathbf{x}}_\mathfrak{h})$ is a slightly modified version of (13), which utilizes the actual scheduled departure time of load l from o, as well as the actual scheduled departure time of the consolidated load from h, instead of the sort level $dep(o_l^\eta)$ and $arr(o_\mathfrak{h}^\eta)$, when checking the time-feasibility at the operational level.

For each $l \in L_C$, let q_l denote the planned volume and Q_l denote the total volume of its trailer type. Define f_l^p as the cost of using the trailer of load l on path p. If p is a direct route, f_l^p represents the transportation cost of the trailer of load l from o_l to d. If p is a consolidation route, f_l^p instead accounts for the transportation cost of the trailer of load l from the consolidation point to d. Additionally, let c_l^p be the detour cost of transporting load l to the consolidation point of path p. If p is a direct route, and thus l in this case is not transported to a consolidation point, then $c_l^p = 0$. The optimization model also needs to reason about trailers and capacities. For that purpose, it is important to introduce some notations for the origin of the last leg of path p. If p is a direct route (o, d), then $oll_p = o$. If p is a consolidation route, then (o, h, d), $oll_p = h$.

2) Optimization Model: For each l and every associated path $p \in P_l$, the optimization model introduces two binary decision variables: • $\xi_l^p \in \{0,1\}$ represents whether load l is assigned to path p; • $\nu_l^p \in \{0,1\}$ represents whether the trailer of l is used for transport on path p. If $\nu_l^p = 0$, l's trailer is eliminated and l is consolidated into another load's trailer. The partial loads consolidation optimization problem is formulated as follows:

$$\min \sum_{l \in L} \sum_{p \in P_l} (c_l^p \xi_l^p + f_l^p \nu_l^p) \tag{15a}$$

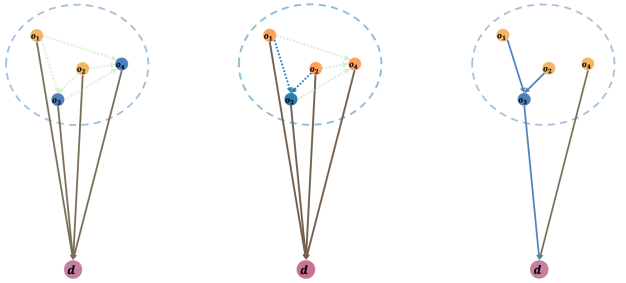
$$\text{s.t.} \sum_{p \in P_l} \xi_l^p = 1, \quad \forall\ l \in L \tag{15b}$$

$$\sum_{l \in L} \sum_{p \in P_l | oll_p = h} q_l\ \xi_l^p \leq \sum_{l \in L} \sum_{p \in P_l | oll_p = h} Q_l\ \nu_l^p, \quad \forall\ h \in H \tag{15c}$$

$$\xi_l^p, \nu_l^p \in \{0,1\}, \nu_l^p \leq \xi_l^p, \quad \forall\ l \in L, \forall\ p \in P_l \tag{15d}$$

Constraints (15b) ensure that each load selects exactly one path. Capacity constraints (15c) guarantee that sufficient trailers are available to transport all consolidated loads at h. Finally, compatibility constraints (15d) ensure that load l can only provide capacity along path p if load l is routed along path p. The objective is to minimize the total transportation and trailer usage costs. Conceptually, this model can be viewed as a generalized assignment problem featuring variable capacity constraints, akin to the network design formulations in [6].

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(a) Feasible Path Generation. (b) Consolidation Paths Selection. (c) Final Consolidation Results

Fig. 4: An Illustration of the Optimization Component.

C. An Illustration of the Optimization Component

Figure 4 presents an example to illustrate each step of the optimization component. Consider four partial loads $l_1, \ldots, l_4 \in L_C$. The corresponding origins $o_1, \ldots, o_4$ have scheduled departure times $t_1 < t_2 < t_3 < t_4$, where o_3 and $o_4 \in H$ represent consolidation points identified by the ML component. In Figure 4a, dark brown arrows show scheduled load transportation from $o_1, \ldots, o_4$ to the common destination d, and light dashed green arrows indicate alternative consolidation paths as described in Section IV-B1. Then, Figures 4b and 4c depict the consolidation decisions made by the optimization in accordance with Section IV-B2. In Figure 4b, the paths $o_1 \rightarrow o_3$ and $o_2 \rightarrow o_3$ are selected. As shown in Figure 4c, the three originally scheduled trailers from o_1, o_2, and o_3 to d are reduced to two better-utilized trailers due to consolidation at o_3, resulting in fewer trailers, less total travel distance, and lower overall transportation costs.

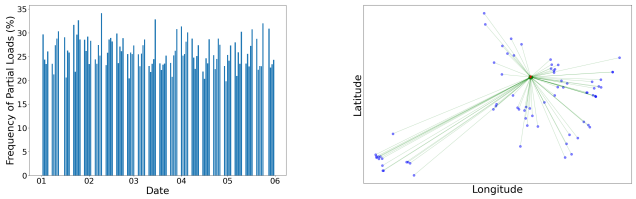
V. Experimental Setting

SPOT was evaluated through extensive experiments conducted on a real-world freight transportation dataset provided by the industrial partner. The experiments evaluate the performance of SPOT against the non-consolidation TruckLoad (TL) load planning currently employed by the industrial partner, and a nearest-neighbor-based heuristic (NNCH) algorithm for load consolidation. The section describes the experimental setting, including an overview of the dataset, the baselines, and the comparison metrics.

A. Datasets

The dataset comprises six months of freight transportation records provided by the industry partner. It contains over two million loads, 39% of which are categorized as partial loads, defined as having a capacity utilization below 80%. The records cover approximately one thousand terminals throughout the United States and include thousands of terminal-and-sort combinations serving as load origins and destinations.

A basic analysis and visualization of historical data revealed potential consolidation opportunities. Figure 5a shows that, for a specific destination, partial loads occur consistently on weekdays, with the daily percentage of partial loads exceeding 25% on average. Moreover, for the same destination on a specific operational day, the spatial distribution of partial loads exhibits clustering characteristics, which can be used to facilitate load consolidation (Figure 5b).



(a) Daily Count of Partial Loads. (b) Spatial Distribution of Partial Loads.

Fig. 5: Daily Count and Spatial Distribution of Partial Loads for a Specific Destination.

In the six-month load dataset, the load data from the final three weeks was utilized as the testing dataset for algorithm comparison. The remaining data was used as training input for the ML component of SPOT. Thirty destinations are selected for experiments exhibiting varying numbers of daily partial load occurrences. Among these, five destinations (**Tier 1**) had the highest daily volume of partial loads, ten destinations (**Tier 2**) had a moderate daily volume, and the remaining fifteen destinations (**Tier 3**) had relatively low levels of daily partial loads. Each destination is evaluated over 15 consecutive weekdays (3 weeks), and the average performance is reported.

B. Baselines

SPOT is compared against two different baselines.

TruckLoad (TL): The TL transportation load planning approach mirrors the current strategy employed by the industrial partner and does not consider load consolidation options. Regardless of its utilization level, each load is directly transported from origin to destination. TL is a baseline in almost every study within this domain [2], [5], [9].

Nearest Neighbor Consolidation Heuristic (NNCH) Algorithm: The NNCH algorithm is introduced as an enhancement of TL, to show the benefits of the SPOT optimization component over a simple informed heuristics. NNCH follows the general principles of the nearest-neighbor heuristic proposed in [5], with necessary modifications to adapt to terminal consolidation. This adaptation is essential because [5] was initially designed for the m-PD-VRPTW formulation. NNCH, detailed in Algorithm 1, operates as follows: for each load, the algorithm iterates through all other loads, starting from the nearest and proceeding to the farthest, and consolidates loads as long as the capacity constraints are satisfied. The input to NNCH matches the input used in SPOT's consolidation optimization, i.e., the set of loads L_C.

C. Metrics

Two types of evaluation are performed to understand the contribution pf SPOT and its components. The first type of evaluation analyzes the effectiveness of the operational consolidation decisions by measuring normalized total travel distance (*Travel Distance (%)*), the percentage reduction in transportation costs (*Cost Reduction (%)*), and the percentage of partial loads cut (*Loads Cut (%)*) for selected operational

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Algorithm 1: Nearest Neighbor Consolidation Heuristic (NNCH)

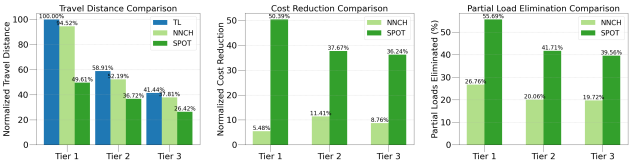
Input: Set of all feasible paths (**paths**), where each path contains origin information $(o, o_quantity, o_capacity, o_departure_time)$, intermediate terminal information $(h, h_quantity, h_capacity)$, and travel time between them $o_h_travel_time$.
Output: List of consolidation decisions (o, h).

```
Sort paths by o_departure_time;
Initialize consolidated_decisions ← [];
while paths is not empty do
    Extract the first path and corresponding o;
    Find candidates C ⊆ paths where o = o and h ≠ o;
    consolidated ← False;
    Sort C by o_h_travel_time (ascending);
    foreach candidate ∈ C do
        Extract (h, h_quantity, h_capacity) from candidate;
        if o_quantity + h_quantity ≤
          max{o_capacity, h_capacity} then
            Add (o, h) to consolidated_decisions;
            Remove paths involving o and h from paths;
            consolidated ← True;
            break;
        end
    end
    if not consolidated then
        Add (o, o) to consolidated_decisions (fallback to TL);
        Remove paths involving o from paths;
    end
end
return consolidated_decisions;
```

days and destinations. These metrics enable a comparative analysis of SPOT against NNCH and TL. The second type of evaluation measures the contributions of the ML component. For consolidation points, the evaluation reports: *Coverage*, the ratio of consolidated partial loads to the total number of partial loads; *CP Ratio*, the proportion of partial load origins serving as consolidation points; and *Daily Loads Per CP*, the average number of loads processed per consolidation point. The evaluation reports route-related metrics including *Path Freq.* that captures the frequency of optimal consolidation routes linked to consolidation points and *Num of Paths* that captures the proportion of feasible routes selected by SPOT. Together, these metrics highlight the effectiveness of the ML component in identifying recurrent consolidation opportunities, narrowing the optimization search space, enhancing consolidation decisions, and offering insights for tactical load planning.

VI. Experimental Results

This section presents the experimental results of SPOT. The primary goal is to address the following research questions: *(Q1) Operational-Level Performance:* Can SPOT provide effective consolidation decisions at an operational level, directly enabling cost reductions in industrial settings? *(Q2) Tactical Insights from Historical Data:* Can SPOT extract meaningful information from historical data that supports load consolidation decisions and offers insights for tactical planning? *(Q3) Computational Efficieny:* Is SPOT sufficient to handle load consolidation across a large-scale transportation network, or is its performance constrained by computational complexity? All experiments were implemented in Python 3.10.15, with Gurobi 12.0.0 employed to solve the optimization problems.



(a) Travel Distance (b) Cost Reduction (c) The Number of Partial Loads Cut

Fig. 6: Consolidation Performance Comparison by Tier.

A. Consolidation Performance

Let ϵ denote the maximum threshold for grouping nodes in the clustering phase, and τ the minimum frequency of patterns in historical data for CFIM. Figures 6 and 7 present results for the configuration $(\epsilon = 0.30, \tau = 5)$, which represents the optimal setting for SPOT. Additional experimental results under various (ϵ, τ) combinations are provided in Tables I and II. Empirically, $\epsilon \in \{0.2, 0.25, 0.3\}$ corresponds to clustering angles of approximately $20^\circ \sim 30^\circ$, while $\tau \in \{5, 10\}$ captures patterns occurring at least monthly or biweekly.

Figure 6 presents a comparison of SPOT's consolidation performance against TL and NNCH across the three destination tiers. By combining the ML component with feasible path generation and optimization, SPOT delivers efficient consolidation outcomes. It consistently outperforms the alternatives across all tiers.

Compared to the existing operational method (TL), SPOT delivers notable reductions in both travel distance and transportation costs. For Tier-1 destinations, which handle high daily volumes of partial loads, SPOT improves performance compared to TL by roughly 50%. Even in Tier-2 and Tier-3 destinations, where consolidation opportunities are limited due to lower volumes, it maintains cost savings of over 36% (Figure 6a). SPOT achieves cost reductions of at least three times those of NNCH, in Tier-2 and up to nine times in Tier-1 (Figure 6b). In terms of the number of loads being cut, the difference between NNCH and SPOT in Loads Cut (%) is relatively small compared to their large gap in cost reduction. Across all three tiers, NNCH cuts about half as many loads as SPOT with significantly lower cost reductions, especially in Tier-1, where the savings are far below half of what SPOT achieves. This highlights NNCH's shortcoming as a greedy heuristic, often making poor consolidation choices. This pattern is further illustrated in Table I, where the best results for TL, NNCH, and SPOT are marked in bold. Interestingly, NNCH's best cost reductions do not align with its highest number of loads cut. In contrast, SPOT consistently shows that smarter consolidation leads to a clear, positive correlation between the number of loads cut and cost savings. The superior consolidation performance of SPOT provides a strong answer to ***(Q1)***, demonstrating its ability to deliver the best operational-level performance.

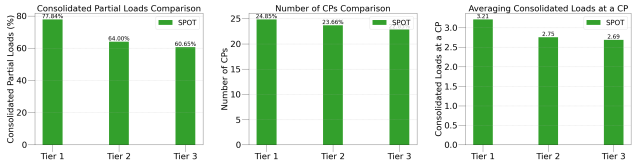
B. Statistics regarding Consolidation Points

Figure 7 demonstrates the strong synergy between the ML and optimization components. When using SPOT, 60%–80% of daily partial loads are consolidated at just 20% of origins,

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	ϵ	τ	Method	Travel Distance (%)	Cost Reduction(%)	Loads Cut (%)
Tier 1	-	-	**TL**	**100.0**	-	-
	0.20	5	NNCH	94.07	5.93	25.38
			SPOT	50.38	49.62	54.94
		10	**NNCH**	**90.08**	**9.92**	**23.09**
			SPOT	54.19	45.81	51.01
	0.25	5	NNCH	94.06	5.94	26.01
			SPOT	50.25	49.75	55.28
		10	NNCH	90.53	9.47	24.05
			SPOT	53.55	46.45	51.59
	0.30	5	NNCH	94.52	5.48	26.76
			SPOT	**49.61**	**50.39**	**55.69**
		10	NNCH	91.25	8.75	25.0
			SPOT	52.97	47.03	52.45
Tier 2	-	-	**TL**	**58.91**	-	-
	0.20	5	NNCH	51.66	12.31	18.06
			SPOT	38.33	34.94	38.82
		10	**NNCH**	**48.82**	**17.13**	**15.66**
			SPOT	41.06	30.3	33.77
	0.25	5	NNCH	52.19	11.41	19.19
			SPOT	37.47	36.4	40.38
		10	NNCH	49.21	16.47	16.67
			SPOT	39.99	32.12	35.48
	0.30	5	NNCH	52.19	11.41	20.06
			SPOT	**36.72**	**37.67**	**41.71**
		10	NNCH	49.56	15.88	17.7
			SPOT	39.16	33.53	37.09
Tier 3	-	-	**TL**	**41.44**	-	-
	0.20	5	NNCH	36.75	11.32	17.25
			SPOT	27.36	33.98	37.14
		10	**NNCH**	**34.06**	**17.82**	**15.43**
			SPOT	28.91	30.24	32.56
	0.25	5	NNCH	37.69	9.06	18.86
			SPOT	26.73	35.5	38.51
		10	NNCH	35.0	15.54	16.75
			SPOT	28.37	31.54	34.07
	0.30	5	NNCH	37.81	8.76	19.72
			SPOT	**26.42**	**36.24**	**39.56**
		10	NNCH	35.3	14.8	17.34
			SPOT	28.14	32.09	34.78

TABLE I: Travel Distance, Cost Reduction, and the Number of Partial Loads Cut.



(a) Cov (b) CP Ratio (c) Loads Per CP

Fig. 7: Consolidation Points Statistics

which act as consolidation points. This low ratio of consolidation points significantly eases tactical-level planning for the industry, as fewer terminals require adjustments, making operational execution of load consolidation more practical. These efforts are well justified, given that most daily partial loads are included in the consolidation process. Besides, as shown in Table II – *Num of Paths*, SPOT considers only around 40%–50% of all time-feasible consolidation paths. Importantly, these selected routes are those that frequently appear in historical records (Table II – *Path Freq*). These facts demonstrate the ML component's ability to recognize repeating patterns and significantly reduce the search space for optimization.

Furthermore, Tables I and II highlight a notable trend in the behavior of the ML module, which offers valuable insights for tactical planning. Specifically, increasing ϵ (loosening clustering constraints) or decreasing τ (lowering the threshold for frequent patterns), or both, leads to greater cost reduction (*Cost Reduction* in Table II) and an increase in *CP Ratio* and *Num of Paths* in Table II. As discussed previously, from an industry perspective, it is crucial to undertake tactical-level preparations

	ϵ	τ	Path Freq (%)	Num of Paths (%)	Coverage (%)	CP Ratio (%)	Daily Loads Per CP
Tier 1	0.20	5	30.25	47.74	76.54	24.34	3.22
		10	31.8	39.19	71.32	23.05	3.2
	0.25	5	30.49	50.71	76.8	24.37	3.23
		10	31.94	43.47	71.99	23.17	3.19
	0.30	5	30.38	53.66	77.84	24.85	3.21
		10	31.87	46.7	73.01	23.41	3.22
Tier 2	0.20	5	25.11	35.18	60.15	22.63	2.7
		10	27.27	29.17	53.05	20.48	2.62
	0.25	5	25.22	38.09	62.19	23.16	2.72
		10	27.07	31.41	55.63	21.15	2.66
	0.30	5	25.36	41.95	64.0	23.66	2.75
		10	27.14	34.64	57.68	21.71	2.71
Tier 3	0.20	5	23.97	40.61	56.67	21.49	2.68
		10	26.23	34.7	50.7	19.56	2.61
	0.25	5	24.2	44.94	59.21	22.42	2.67
		10	26.56	38.17	53.16	20.6	2.59
	0.30	5	24.34	47.43	60.65	22.85	2.69
		10	26.66	39.96	54.23	20.91	2.61

TABLE II: Consolidation Points & Paths Statistics of SPOT.

by coordinating the load consolidation planning with other components of the transportation management system, such as driver scheduling, route assignments, equipment availability, and intermediate terminal arrangements. In the experiments, the associated preparation costs are proportional to *CP Ratio*, the number of paths requiring modifications or adjustments at the tactical level to ensure that the chosen consolidation routes are operationally feasible and achieve cost reductions. The associated preparation costs increase with the *CP Ratio* and the *Num of Paths*, as both metrics reflect how many terminal and route candidates may require tactical adjustments to ensure that the selected consolidation plans are feasible in practice. Consequently, SPOT provides decision makers with a clear trade-off between the operational cost of advanced effort (i.e., how many candidates to prepare for) and the potential transportation cost savings. Decision makers can make informed choices by examining the exact paths identified by the ML component and their corresponding performance at both the tactical and operational levels.

In summary, the observed consolidation performance supports the conclusion that SPOT effectively extracts critical insights from historical data at the tactical level and can develop effective, detailed consolidation decisions at the operational level, thereby addressing questions *Q1* and *Q2*.

C. Computational Efficieny

SPOT considers each destination independently, offering two primary advantages when scaling to the entire transportation network. First, parallelizing the computations across multiple destinations is straightforward because each destination is computed independently, resulting in high overall efficiency. Second, restricting the destination-based consolidation leads to a binary optimization model that can be solved to optimality quickly. For Tier 1 destinations, SPOT requires an average of 18 seconds of computation time while, for Tiers 2 and 3, it only requires 5 seconds and 2 seconds, respectively, demonstrating its ability to handle network-wide operational scenarios efficiently. These results demonstrate SPOT's capacity to efficiently handle network-wide operational scenarios, thereby addressing *Q3* as well.

VII. Conclusion

This study introduces SPOT, a novel framework for load consolidation that integrates machine learning techniques with

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optimization to improve load consolidation. The ML component combines spatio-temporal clustering with constrained frequent itemset mining (CFIM), while the optimization component employs a MIP model to ensure feasible and cost-effective decisions. By bridging tactical insights with operational constraints, SPOT, not only provides actionable guidance at the tactical level, but also delivers efficient consolidation decisions to guide route selection on operational days. Extensive experiments on real-world load data demonstrate SPOT's effectiveness, showing consistent and substantial cost reductions compared to baseline methods. SPOT also serves as a blueprint for further research on combining ML and optimization models for logistics and supply chain applications, underscoring the benefits of leveraging historical data in today's era of data abundance.

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